Programming Assignment 1: Cooperative Member Ledger

Data Structures & Algorithms Laboratory Handout

Instructor: Bavith

Problem Overview

A cooperative society maintains a digital ledger of its registered members. Each member is uniquely identified and holds a specific amount of share capital, along with address details comprising a village and a pincode.

Your task is to design a C program utilizing nested structures to capture, compute, and report membership details as specified below.

> **Input Validation Rule:** The total number of members `N` must be greater than `0` and less than or equal to `200`. Any count outside this valid range must immediately output `INVALID COUNT` and terminate execution.

Data Structure Requirements

Address Structure

- `village`: Character array / string
- `pincode`: Integer value

Member Structure

- `id`: Integer member ID
- `name`: Character array / string
- `share_capital`: Floating-point value
- Nested `Address address`

Task Specifications

- Read `N` members into an array of structures from standard input.
- Print a formatted ledger listing all members in entry order.
- Compute and output the total accumulated share capital across all members.
- Identify and reprint the full details of the member holding the highest share capital (note: in case of a tie, preserve the first incumbent encountered).

Sample Test Cases

#	Input (stdin)	Expected Output (stdout)
1	4 1041 Basavaraj 25000.00 Hirekerur 581111 1042 Shantamma 41500.50 Ranebennur 581115 1043 Mallikarjun 18750.25 Byadgi 581106 1044 Girija 41500.50 Haveri 581110	MEMBER LEDGER 1041 Basavaraj Hirekerur 581111 25000.00 1042 Shantamma Ranebennur 581115 41500.50 1043 Mallikarjun Byadgi 581106 18750.25 1044 Girija Haveri 581110 41500.50 Total share capital (Rs): 126751.25 Highest holder: 1042 Shantamma Ranebennur 581115 41500.50
2	1 2201 Ningappa 9000.00 Savanur 581118	MEMBER LEDGER 2201 Ningappa Savanur 581118 9000.00 Total share capital (Rs): 9000.00 Highest holder: 2201 Ningappa Savanur 581118 9000.00
3	0	INVALID COUNT

Submission Guidelines

Ensure your program strictly matches the formatting guidelines and output strings shown in the test cases above. Solutions will be evaluated automatically against standard test suites.

> **Instructor Note:** Complete your implementation and submit the source file via the portal before the deadline. Reach out during lab hours for clarification.
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> **— Bavith**